

School of Computing and Informatics

Website Design and Development

Test Plan

Student name: Mohammad Aldeghemat

Test Plan

1. Scope (in scope, out scope)

For Takke, I am creating a learning platform that users can browse courses and buy them, and it includes sign-in and sign-up process, it offers a comprehensive learning trip for the students.

In Scope:

- Sign-in/Sign-up: the users will register and create an account, then log-in to their account using the information they used in the sign-up. The username is unique to identify which user added courses to cart, it is to separate each user's cart.
- CRUD Functionality:
  - Create: users can sign-up, add to cart.
  - Read: the page reads courses, my courses, and my profile details then display them.
  - Update: users can update their profile info such as email and password but not the username.
  - Delete: users can remove course from the cart.
- Profile update: users can update their profiles such as email and password and the changes is updated after submitting.
- Responsiveness: Takke will work well on different devices screen sizes and include flexible layout.

Out Scope:

- Payment: There is no real checkout, when the user clicks checkout, it redirects him to my courses page, and it displays the courses that are in the cart.
- Security: encryption, hashed passwords, and multi factor authentication are not included on the website.
- Search Functionality: Takke will not provide a search bar that searches for courses.
- Custom theme: The user will not be able to customize theme of website like (light, dark).

In conclusion, the scope clearly identify Takke's features and what will be in the website and what will not be in the website, and this make sure that everything is clear for the developer, designer, and stakeholders.

2. Schedule

No	Date from	Date to	Number of hours	Test Type *(Responsiveness, form validation, consistency … etc.)*	Description
1	25/12/2024	26/12/2024	4 hours	sign in/sign up validation	Test the sign in/signup form such as (duplicate user or email, password length, empty fields invalid username or password) and if it really create account.
2	26/12/2024	28/12/2024	1.5 hours	Content validation	Test the content that appear to the logged in and logged out user.
3	30/12/2024	30/12/2024	3 hours	Links validation	Test all links and they must open including the pages.

4	8/1/2025	8/1/2025	1 hour	Icons validation	Test all icons that have a link and if they appear correctly in the selected pages.
5	11/1/2025	16/1/2025	6 hours	Functionality	Test the functionalities like (add to cart, checkout).
6	12/1/2025	12/1/2025	3 hours	Responsiveness and design	Check that all designs work correctly, and the pages work well on all devices screen sizes.

3. Test Cases (at least 25 Test cases)

No	Description	Type *(Responsive, Navigation, look and feel, page titles, form validation ..etc.)*	Steps	Expected results	status
1	Test the sign-up form for duplicate username or email.	sign in/sign up validation	1.Open the sign-up page. 2. enter an existing username or email. 3. submit the form.	A java script alert should appear saying “Failed, username and email should be unique” and refill the form	Pass
2	Test the sign-up form if the password is less than 8 characters.	sign in/sign up validation	1.Open the sign-up page. 2. enter a password less than 8 characters. 3. submit the form.	A warning should appear and not continue with the sign-up process and refill the password	Pass
3	Test the sign in by filling non-existence username or password.	sign in/sign up validation	1. open the sign on page. 2. enter a non-existent username or password. 3. submit the form	A java script alert should appear saying “Invalid username or password” and refill the form	Pass
4	Test the sign in form if the password is less than 8 characters.	sign in/sign up validation	1.Open the sign in page. 2. enter a password less than 8 characters. 3. submit the form.	A warning should appear and not continue with the sign in process and refill the password	pass
5	Test the home page content for logged out user (it should be different than logged in user).	Content validation	1. open the home page without signing in. 2. see the home page result.	For the logged-out users the content that should not appear is (logout link, my profile page link, my cart page link) and only 4 courses appear, sign in, and sign-	Pass

				up link should appear.	
6	Test the home page content for logged in user (it should be different than logged in user).	Content validation	1. sign in and open the home page. 2.see the home page result. 3. compare between the two results.	For logged in users, the content that appears is different like (logout link, my courses link, my cart link, my profile link) and all courses should appear. The content that shouldn't appear is the sign in and signup link	Pass
7	Test the links for logged in users.	Links validation	1. sign in and open the home page. 2. open all links in the page.	All links should work correctly and go the page successfully, for example: my courses link → my courses page, My cart links → my cart page	Pass
8	Test the links for logged out users.	Links validation	1. open the home page. 2. open all links in the page.	All links should work correctly and go the page successfully, for example: my courses link → sign in page, Sign up link → Sign up page	Pass
9	Test the footer links.	Links validation	1. open the home page. 2. Click on the links in the footer.	All links should work correctly and go the page successfully, LinkedIn link → LinkedIn profile page, Email → open the email app and compose a message to the selected email	Pass

10	Test if the sign-up page work correctly and it creates an account.	sign in/sign up validation	1. open the sign-up page. 2. fill in the form and submit. 3. open the database and see, is the data inserted successfully?	The data should be inserted to the database	Pass
11	Test if the sign in page work correctly and it starts a session.	sign in/sign up validation	1. open the sign in page. 2. fill in the form and submit. 3. you will be redirected to the home page. 4. add anything to the cart and go to my cart page. 4. e text will be shown saying “username cart”.	The user cart should appear in it a message, for example the username is moe, the message will be “moe cart”	Pass
12	Test the icons that appear in pages like (sign in, signup, my cart, my profile pages).	Icons validation	1. open each page. 2. see if the home icons are in all these pages, and the profile icon in my cart pages.	The icons should be displayed and if they act as a link the link work correctly	Pass
13	Test the sign up by putting only username and password without the email.	sign in/sign up validation	1. open sign-up page. 2. fill in username and password. 3. press submit.	A message will show on the email field “Please fill out this field”	Pass
14	Test the sign up by putting only email and password without the username.	sign in/sign up validation	1. open sign-up page. 2. fill in email and password. 3. press submit.	A message will show on the email field “Please fill out this field”	Pass
15	Test the sign I by putting only username without the password.	sign in/sign up validation	1. open the sign in page. 2. fill in the username field. 3. press submit.	A message will show on the email field “Please fill out this field”	Pass
16	Test the sign I by putting only password without the username.	sign in/sign up validation	1. open the sign in page. 2. fill in the password field. 3. press submit.	A message will show on the email field “Please fill out this field”	Pass
17	Test add to cart button if it adds it to the cart table.	Functionality	1. sign in. 2. open the home page and go to courses.	Data should be inserted the cart table in data base	Pass

			3. press add to cart on any course.	for the specific user.	
18	Test add to cart button if I add the same course multiple times.	Functionality	1. sign in. 2. open the home page and go to courses. 3. press add to cart on any course and press it again.	It should appear a message “Course Is Already Added”.	Pass
19	Test the cart page if it shows the courses that are in the cart table.	Functionality	1. sign in. 2. add course to cart. 3. go to cart.	The cart should show the courses that are added.	Pass
20	Test the cart page if it calculates the total price right.	Functionality	1. add multiple courses. 2. go to cart page and see if the total is right.	The total should be printed right.	Pass
21	Test the logout link if it terminates the user session.	Functionality	1. sign in. 2. go to home page and press on logout.	Terminates the session, and show the page for logged out users	Pass
22	Test the cart for multiple users, does it specify which cart is for which user?	Functionality	1. sign up and make two accounts. 2. add multiple courses for the first user and for the second user. 3. compare the cart.	Every user should his specified cart.	Pass
23	Test the checkout functionality in cart page.	Functionality	1. add courses to cart. 2. press on checkout.	It should complete the payment and insert cart data to the owned courses table but Because I didn’t handle payment the checkout will open my courses page that takes the courses data from the cart table.	Fail
24	Check if the user updates his profile it updates on the database	Functionality	1. sign in and go to my profile page. 2. change the email and password	The data should be updated in the database for this user.	
25	Check if the user delete course from cart it deletes from database	Functionality	1. add course to cart. 2. go to my cart page. 3. press on delete for specific course.	The course should be deleted in the database for this user.	Pass

26	Check if the home pages read the courses from the database correctly	Functionality	1. go to the home page. 2. scroll down to courses.	The courses are displayed on the web page, and they were read from the database.	Pass
27	Check on the design of pages if it works correctly.	Responsiveness and design	1. open every page. 2. compare the design to the CSS code.	The design should be working correctly.	Pass
28	Check on the pages if they work on different devices screen sizes.	Responsiveness and design	1. open every page. 2. press inspect. 3. see the website on multiple devices.	The page should work correctly at all screen sizes.	Pass
29	Check if the design as exactly as the wireframe.	Responsiveness and design	1. open every page. 2. open the wireframe. 3. compare every page with the wireframe.	Every page should match the wireframe.	Pass
30	Check if the font is the same in all pages	Responsiveness and design	1. open every page. 2. compare the font in the pages to the font used.	Every page should have the same font(consistency).	Pass

4. Evaluate the Quality Assurance (QA) process

4.1 Requirements

Requirements are important to identify the features of the website, here is a detailed list of achieved and not achieved requirements:

Achieved Requirements:

- User Account Management:
    1. Sign-up: The users can create an account, store their data in the database, and sign in with these data.
    2. Sign-in: users can establish a session by signing in.
    3. Sign-out: users can terminate the session and sign out from their account.
    4. Update User Information: users can update their data in the database such as email and password.

  It was very crucial to achieve the user account management requirements because the sign-up is the core functionality of the website, users need to create accounts and sign in to establish a session and to make their progress in the website private for each users, sign-out was implemented because it is a security that allows user to terminate their session and log out from the account. Also, updating information is implemented to improve user experience, it maintains up to date data. If this requirement is not achieved, then the website is not useful.

- Dynamic 5-6 webpages: My website included CRUD operations:
    1. Read: users can see the courses displayed in home inside the courses section, these courses are retrieved from the database.

2. Insert: users can insert data to the database by creating accounts and adding courses to cart.
3. Update: users can update their information in the database like email and password.
4. Delete: users can delete a course from the cart and database.

It was a big very crucial to achieve this requirement, it reduces the lines of code and reducing file size, by reading data from the database users can see available courses and this ensure that users can see the courses, if it wasn't implemented, we may add courses manually in code, and when a new course added we have to add it in the code which is a waste of time. Also, inserting data to the database is important for data management, the users can create accounts, and add courses to cart, and without this, users can't have a private experience. Updating and deleting data allows users to manage their account and their cart, without this, users can't change their decisions for the courses selected.

- Responsive Design: I created Takke not to only open it from the desktop, but users can open Takke from different device's screen sizes and the website will be displayed well on these different devices. Also, I used flex box and media queries to ensure the responsive design.
  Due to the technological revolution, we have now a lot of devices with a lot of screen sizes, a responsive design was implemented to make sure that Takke works on different screen devices correctly, this improves usability and consistency.

Not Achieved Requirements:

- Checkout: The checkout process is not achieved in the website, instead, I made an idea that if the user press on checkout it will go to my courses page that reads data from the cart table.
- Custom themes: The custom themes that users choose such as light, and dark webpage is not achieved because my website only have a design of a single navy theme.
- Some security measures: encryption, hashing, and multi factor authentication are not achieved because these are advanced, and I will include them in future.

In conclusion, achieved requirements are the core of Takke's functionalities, but not achieved requirements were not achieved due to complexity or being outside the scope.

4.2 Design

Takke's requirements efficiently reflected on the design of Takke, all requirements are converted into a visual, and user-friendly website, and this is how these requirements reflected on the design:

- User Account Management: the features of this requirement were in a clean, and user-friendly form, or button as the logout. The design included labels, input, and input validation, and I make sure that users can easily know the way of how to create an account, login to their account, update their profile information's, and logout from their account. Unlike the sign-in, sign-up, and updating information that were designed as a form, the logout was designed as a button in the nav bar, it is just a button that user presses to log out from their account.

- Dynamic 5-6 webpages: the requirements of the webpages reflected the design in multiple ways, the home page for logged in user is have some different content than logged out users and this attract user to create an account on takke and see the hidden content. Also, the courses in the home page have an add to cart button with an effect to get user attention on it, it makes the users add courses to their cart. For the profile page it was created in a clean, and user-friendly form which I discussed above. But the delete button in the cart page will be besides each course to remove unwanted courses, it was implemented as a button with an effect.
- Responsiveness: the responsiveness requirements reflected the design by using flex box system, and media queries, and I make sure that the website looks great on all devices, and this ensure the user experience and make the website accessible and user-friendly.

Of course, all these requirements we know reflected the design according to the test that I did earlier, the design was tested to make sure that we achieved all requirements. The requirements effectively reflected on the design process, the requirements do not only reflect on the design, but also, they were all met in the website design.

The color palette used was: https://colorhunt.co/palette/f9f7f7dbe2ef3f72af112d4e

4.3 Front-end Development

There was a lot of test cases that were related to the front-end development such as:

- Test Case 2: password input validation.
- Test Case 4: sign-in, test the password if it's less than 8 characters.
- Test Case 5: Test home page content for logged-out users.
- Test Case 6: Test home page content for logged-in users.
- Test Case 7: test the links for logged-in users.
- Test Case 8: test the links for logged-out users.
- Test Case 9: test footer links.
- Test Case 12: test the icons in multiple pages.
- Test Case 13: sign-up, input validation by filling username and password without email.
- Test Case 14: sign-up, input validation by filling email and password without username.
- Test Case 15: sign-in, filling in only username.
- Test Case 16: sign-in, filling in only password.
- Test Case 27: check if the designs work correctly.
- Test Case 28: check if page works in different screen sizes.
- Test Case 29: check if the designs are exactly as the wireframe.
- Test Case 30: check if font is the same in all pages.

4.4 Back-end Development

There was a lot of test cases that were related to the Back-end development such as:

- Test Case 1: sign-up form, testing duplicate username or email.
- Test Case 3: sign-in form, testing if user fill non-existent username or password
- Test Case 10: sign-up, test if it creates an account and insert it into database.
- Test Case 11: sign-in, test if its login and start a session.

- Test Case 17: test add to cart button if it inserts into cart table in database.
- Test Case 18: test add to cart button if the user adds the same course multiple times.
- Test Case 19: test if cart page shows courses that are in cart table.
- Test Case 20: test if the total price of cart calculated right.
- Test Case 21: test logout link if it terminates session.
- Test Case 22: test if the cart specifies cart for every user.
- Test Case 23: test checkout functionality.
- Test Case 24: test if user update his profile, it updates in the database.
- Test Case 25: test if the user can delete a course in cart in database.
- Test Case 26: check if home page reads courses correctly.

All test cases passed the test unless one test case which is test case 23, it failed.

14 test cases are related to the back end, and 16 test cases are related to the front-end development.

5. Critical evaluation of the results of your Test Plan.

Takke test plan was implemented, and I did it in a clear way, and the results provide details of what the website will do including functionality, performance, etc. I did the testing based on the requirements and they made sure that every requirement was implemented.

The test plan had 30 test cases, 29 test cases were passed and 1 test case was failed which is test case 23, so the success rate here is 96.7% and the failure rate is 3.3%. The success rate is high which point that most of the website functionalities, content, design, website validation, and the requirements were implemented correctly. The failure test case was because the case is out of the website scope which is not important for now.

Test plan has a major success including user account management, CRUD operations, Responsiveness and design, Front-end validation, back-end validation, database operations, dynamic webpages, and icons and links.

As there was a major success, there is a major failure also, including the checkout in cart functionality, some security measures, search functionality, and custom themes like light or dark.

Also, there were minor issues that we didn't implement in the test plan or in the website such as Wishlist, user rating, performance optimization.

All of the major and minor failures and issues should be considered to add them in the website in the future.

In conclusion, Takke has strengths and weaknesses, but the strengths are more than the weaknesses, we have 96.7% success rate that overwhelm the failure rate. But improvements such as the checkout system, etc. should be considered to have a successful rate of 100% for Takke to make sure that it remains competitive in its field.

6. Review of the overall success areas of the multipage website.

Takke has a lot of strengths and a lot of areas that are successful, and they point to the success of the implementation of all functionalities, when I did the testing, I saw that most

of the test cases were successful, for the user account management, users can create accounts, sign-in and start a session with their account, logout and terminate their session, and they can update their account details that changes in the database, this aspect is a crucial part of Takke, and it is a success that should never fail.

Another success area is the CRUD operations, theses operations were a major requirement on the website which I implemented correctly in Takke. Users can add courses to their cart which adds the courses to the cart table in the database, users can view courses, their profile, and their cart which they are read from the database, users can update their profile details which updates it in the database, and finally they can delete a course from their cart which deletes it from the database. Without the CRUD operations, and the user account management the website will be useless, CRUD was very crucial aspect to implement and to work correctly, it Is a success area that we can't give it up.

Also, the responsiveness and design of Takke were a major success aspect of the website, Takke can work well on different devices including laptops, Pc's, tablets, and mobile phones. The design matches the wireframe, and the color-palette was chosen effectively. Success of responsiveness and design makes sure that the website is usable on every device not only on Pc's.

The dynamic webpages is another aspect of the success area, you can navigate easily with multiple pages, and the content for logged in users is different than the content for logged out users which provides an efficient accessibility, also, it attract users to create accounts to see the hidden content.

There are many success area of the website such as front-end and back-end validation, everything works correctly including content displayed or the operations that happens in the back, and database operations were correct, it handles many request such as insert, update, read, and delete, also, the icons and links work correctly and they provide a smooth navigation between pages. The success area shows that Takke has many strengths, and it provided a brief information about what can we improve.

7. Justified recommendations for areas that require improvement.

While Takke has many strengths, there are areas that require improvement to improve the overall functionality of the website. The crucial and most important improvement is implementing a checkout system that allows users to complete their purchase, the current checkout button only redirects the user to my courses page which is not a real way of checkout, it is essential to add a checkout process for making Takke functionality to the max level.

Also, security is an area that requires improvements such is implement password hashing, encrypt the data, and making a multi-factor authentication. Also, I should implement a system if the user forgot the password to send him an email to reset the password, this reduces the risk of unauthorized access and makes sure that all data are secured.

Another area that requires improvement is the implementing course progress that tracks users progress for each course, it makes the users learning trip more efficient and it maintain users engagement.

Also, a crucial area that require improvement is adding course details pages, for every course there is a page that displays the content of this course and implement user reviews and ratings for courses which is essential for students to complete their learning journey.

We also should provide a roadmap for each course, and a roadmap for learning sound engineering from beginner to professional and mastering sound engineering.

There are multiple areas that needs improvement such as the search functionality to allow users to search for courses, custom themes like light or dark mode, a Wishlist features to make users save courses for future purchase, adds certificates for each user that finishes a course, and finally adds assignments to users to make the learning journey more engaged.

In conclusion, Takke has a strong successful features, and some areas that needs to improve, by implementing all of these areas of improvement, Takke can be more competitive in its field and it make sure that the website operations are effective, and pointing to a future success.